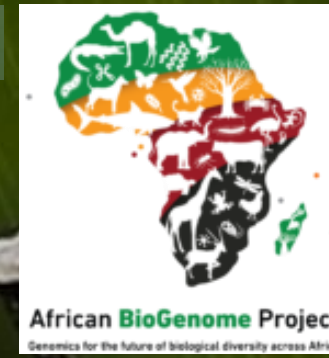




Project Psyche



CREATING A NEW FOUNDATION FOR BIOLOGY

Sequencing Life for the Future of Life



Project Psyche – sequencing and analysing all 11,000 European Lepidoptera



Why Lepidoptera?



Psyche Hubs of Phase 1

★ 1 sequencing hub



Charlotte Wright



Mark Blaxter



Joana Meier



★ 7 sampling hubs



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Muzeul Național de
Istorie Naturală
Grigore Antipa



UNIVERSITY OF OULU



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EVOLUTIVA



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upf.



LUND
UNIVERSITY



Project Psyche community

139 members

from >100 institutions

from 28 countries



Funded by the Horizon 2020 Framework Programme
of the European Union

**10k Lep
Genomes**



Openly published genomes and Genome Notes: Collector(s) as first author(s)

<https://www.darwintreeoflife.org/genomes/genome-notes>

Wellcome Open Research

Wellcome Open Research 2023, 8:278 Last updated: 18 SEP 2023



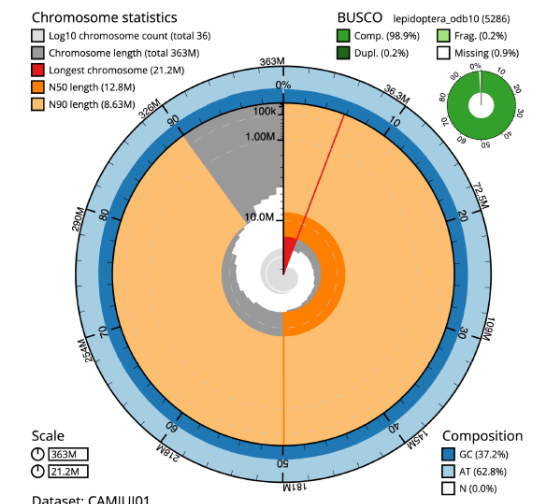
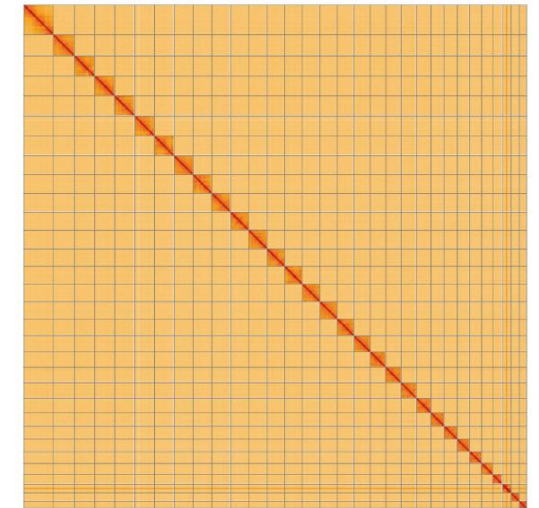
DATA NOTE

The genome sequence of the Dark Spectacle, *Abrostola triplasia* (Linnaeus, 1758) [version 1; peer review: 2 approved]

Douglas Boyes¹⁺, Owen T. Lewis ²,
University of Oxford and Wytham Woods Genome Acquisition Lab,
Darwin Tree of Life Barcoding collective,
Wellcome Sanger Institute Tree of Life programme,
Wellcome Sanger Institute Scientific Operations: DNA Pipelines collective,
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Psyche 1000 !

Trends in Ecology & Evolution



Feature Review

Project Psyche: reference genomes for all Lepidoptera in Europe

Charlotte J. Wright^{1,*}, Niklas Wahlberg², Roger Vila³, Marko Mutanen⁴, Pável Matos-Maraví⁵, Kay Lucek⁶, Irena Kleckova⁵, Leonardo Dapporto⁷, Vlad Dincă⁸, Claudia Bruschini⁷, Christopher W. Wheat⁹, Marta Vila¹⁰, Laura Torrado-Blanco^{10,11}, Valentina Todisco¹², Michal Rindos⁵, Petr Nguyen^{5,13}, Peter O. Mulhair¹⁴, Stefaniya Kamenova¹⁵, Marcus Hicks¹⁶, Marianne Espeland¹⁷, Ines A. Drinnenberg¹⁸, Mónica Doblas-Bajo¹¹, Richard I. Bailey¹⁹, The Project Psyche Community^{20,22} Mark Blaxter¹, and Joana I. Meier^{1,21,*}

Project Psyche is a transnational initiative to generate and study chromosome-level reference genomes of all ~11 000 species of Lepidoptera (butterflies and moths) found in Europe. Here, we describe the decentralised network of collection

Highlights

Lepidoptera are key to ecosystem functioning as herbivores, pollinators, and

Papers planned for the first 1000 genomes

- Chromosome evolution
- Rates of hybridisation across Lepidoptera
- Sex chromosome evolution
- Gene family expansions
- transposable elements
- Cobionts
- Correlation of any of these with speciation rates

and many more

Genómica de los Lepidópteros Neotropicales



https://docs.google.com/forms/d/1FgAfO_ZQtvOKGEWU5bYpxXqFuLL_4feq6TPWqg7gnJQ/edit?pli=1



Dr. Vicencio Oostra
Queen Mary University of London
Dr. Nicol Rueda
Wellcome Sanger Institute



Rede Latinoamericana de Genomica
da Biodiversidade

Red Latinoamericana de Genomica de
la Biodiversidad



Goals

- Form a network to accelerate the sequencing of Biodiversity in Latin America, including countries representatives and large local genome projects
- Create synergies, increase funding opportunities, and solve issues that are common across the region
- Become affiliated to the Earth BioGenome Project and possibly a regional node

First Steps

- Kick Start: Symposium at the Congreso Latinoamericano de Evolución (II CLEVOL) in July 2025
 - gathered projects and countries in Latin America (Brazil - GBB and GenoTropics, Chile - 1000 Genomas, Colombia - group of researchers)
 - informal meetings during Clevol led to the agreement of the Network, where:
 - A series of meetings should take place until the end 2025
 - Each meeting is organised by a different country/project
 - The meetings should attract new people and new countries before agreeing on a structure for the network

First Meetings

- Meeting 1 August (Chile):
 - Presentation of ongoing projects and situations within countries (Chile: 1000 genomas, Brasil: GBB, Genotropics, Colombia, Argentina)
 - Creation of a shared drive for regional coordination.
 - Creation of a list of available courses and interested researchers by country
- Meeting 2 September (Brasil/GenoTropics):
 - Discussion about the EBP Phase II paper and how it fits the reality in Latin America
 - Creation of a list of laboratories in Latin America with ONT technology (in the process of) being used for reference genomes
- Meeting 3 October (Argentina):
 - Presentation of the current situation in Argentina, and the group of researchers forming the Argentinian node of the network
 - Upcoming Events: proposal of a symposium at the Latin American Genetics Conference in 2026; Pore-C course organised by Chile; Discussion with BGA for training in Latin America

Next meeting: organised by Colombia

Group of representatives (under construction)

- Juliana Vianna, Universidad Andrés Bello (Chile)
- Mariana Telles, Federal University of Goiás & PUC-Goiás (Brasil)
- Alexandre Aleixo, Vale Technology Institute (Brasil)
- Laura Kamenetzky, University of Buenos Aires (Argentina)

Supported by: Camila Mazzoni, Leibniz Institute for Zoo and Wildlife Research (Europe/Brasil)